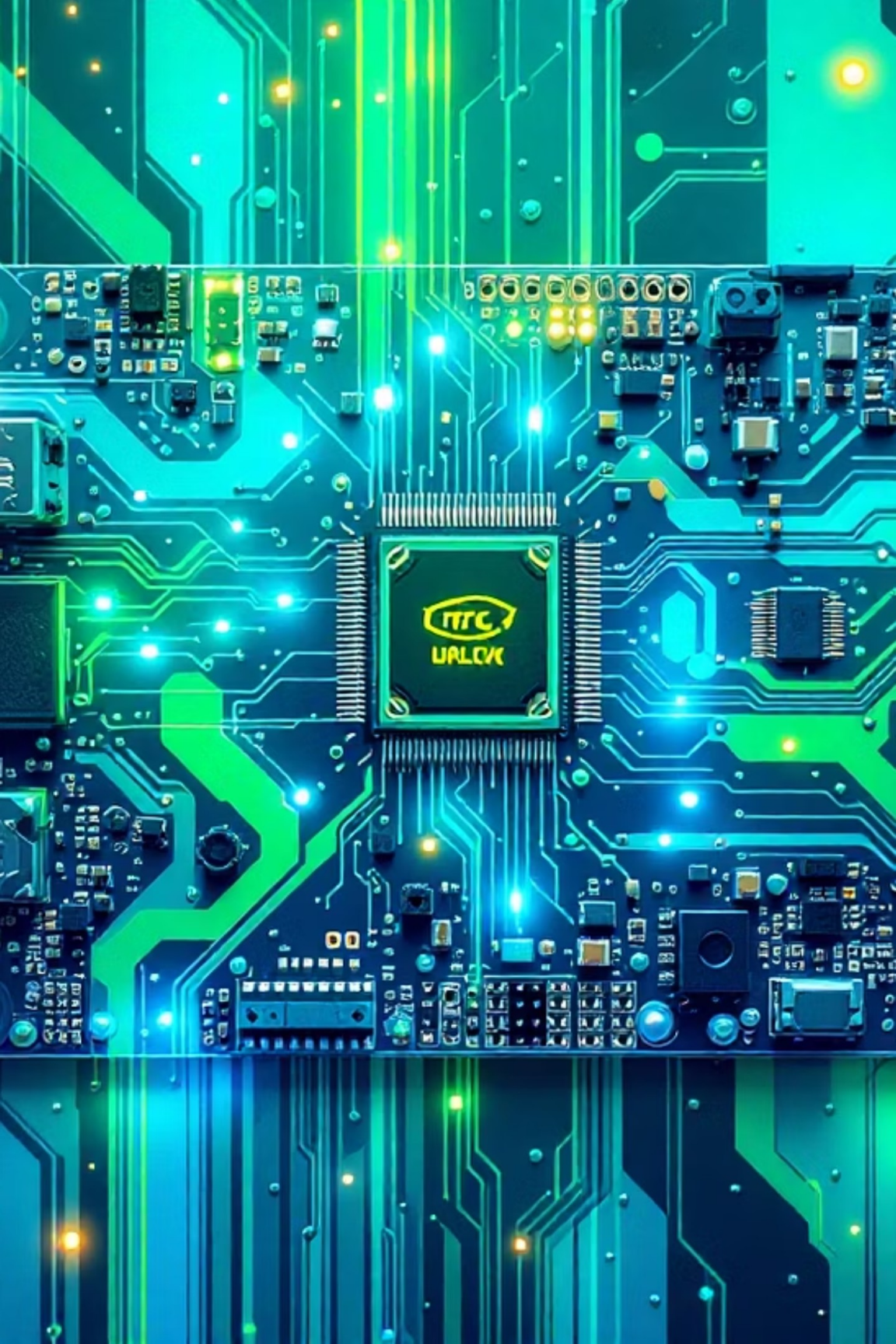




Zephyr Workshop – Kconfig + DeviceTree

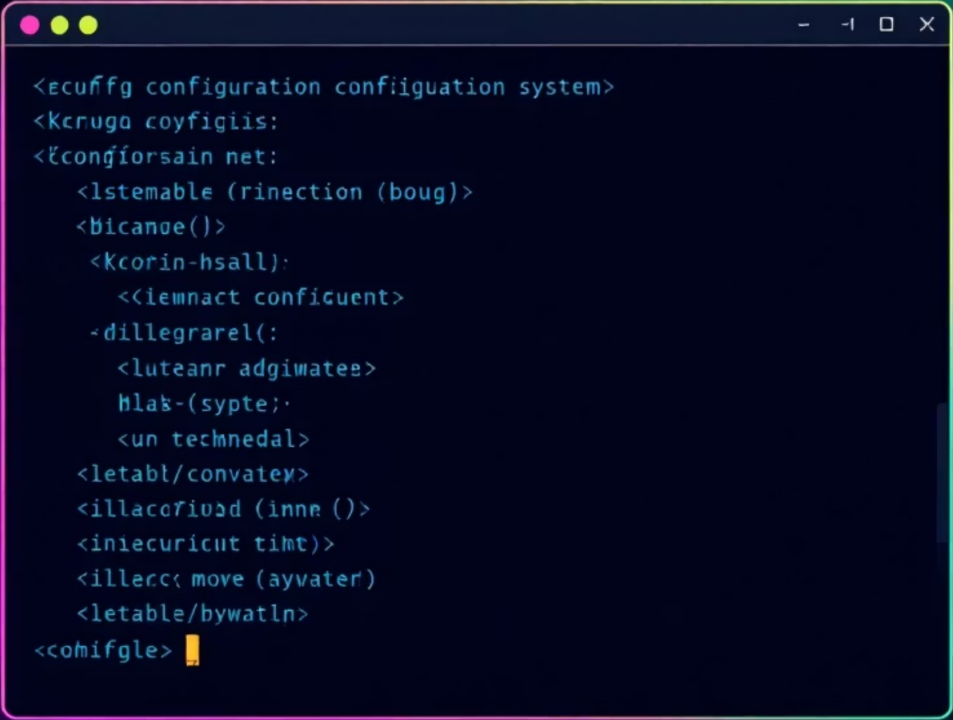
Understanding how Zephyr RTOS configures features and hardware

What is Kconfig?

Configuration system for:

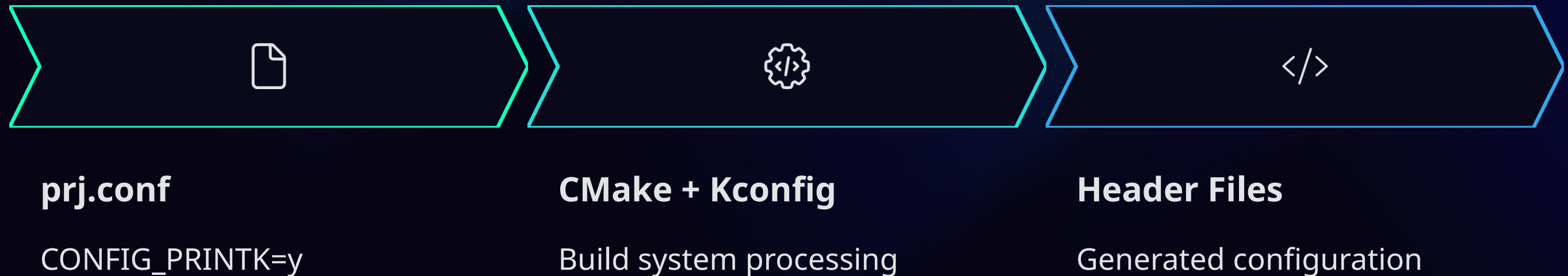
- System features
- Drivers
- RTOS options

Options start with 'CONFIG_'



```
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  <bicanoe()>
  <kcorin-hsall):
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  <iniecuricut timt>
  <illercc move (ayvater)
  <letable/bywatln>
<omifgle> █
```

Kconfig – How it Works

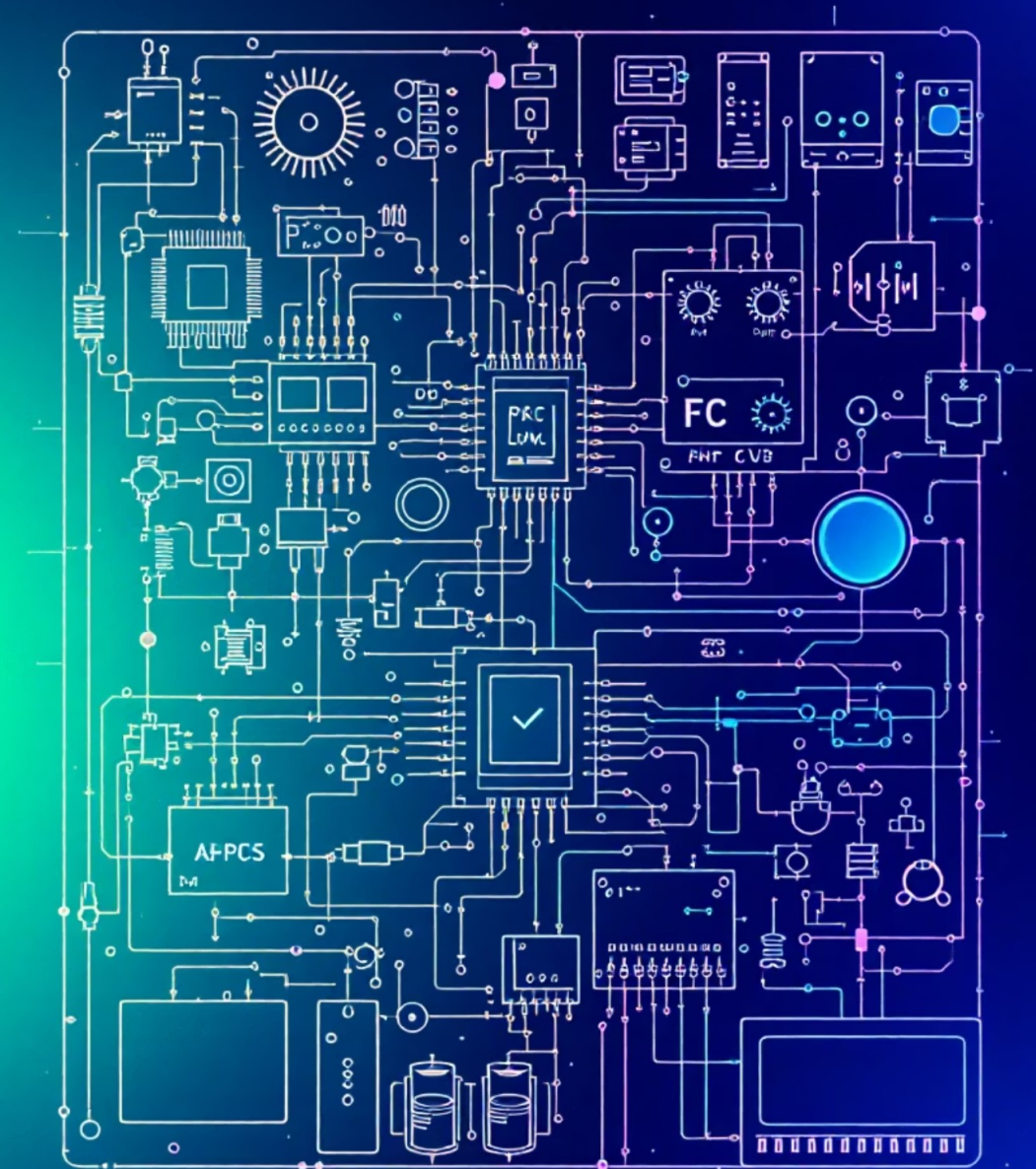


Use 'west build -t menuconfig' to explore

What is DeviceTree?

Hardware description language that:

- Describes hardware layout and connections
- Used by Zephyr at build time to generate hardware abstraction code.
- Files: *.dts (base), *.overlay (app-level override)



DeviceTree

Nodes

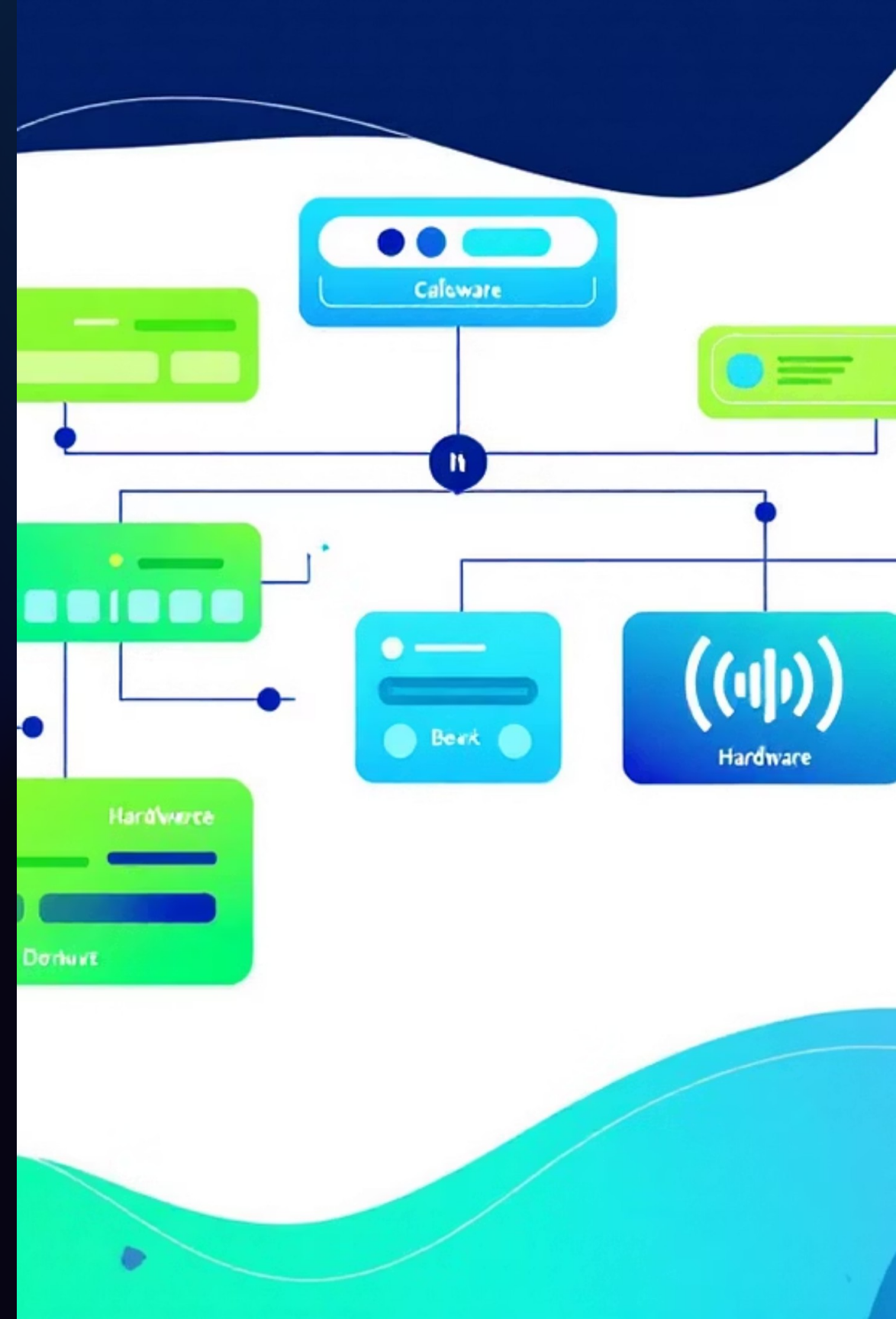
Hardware blocks (e.g.,
&gpio0)

Properties

Configuration data (pins,
levels)

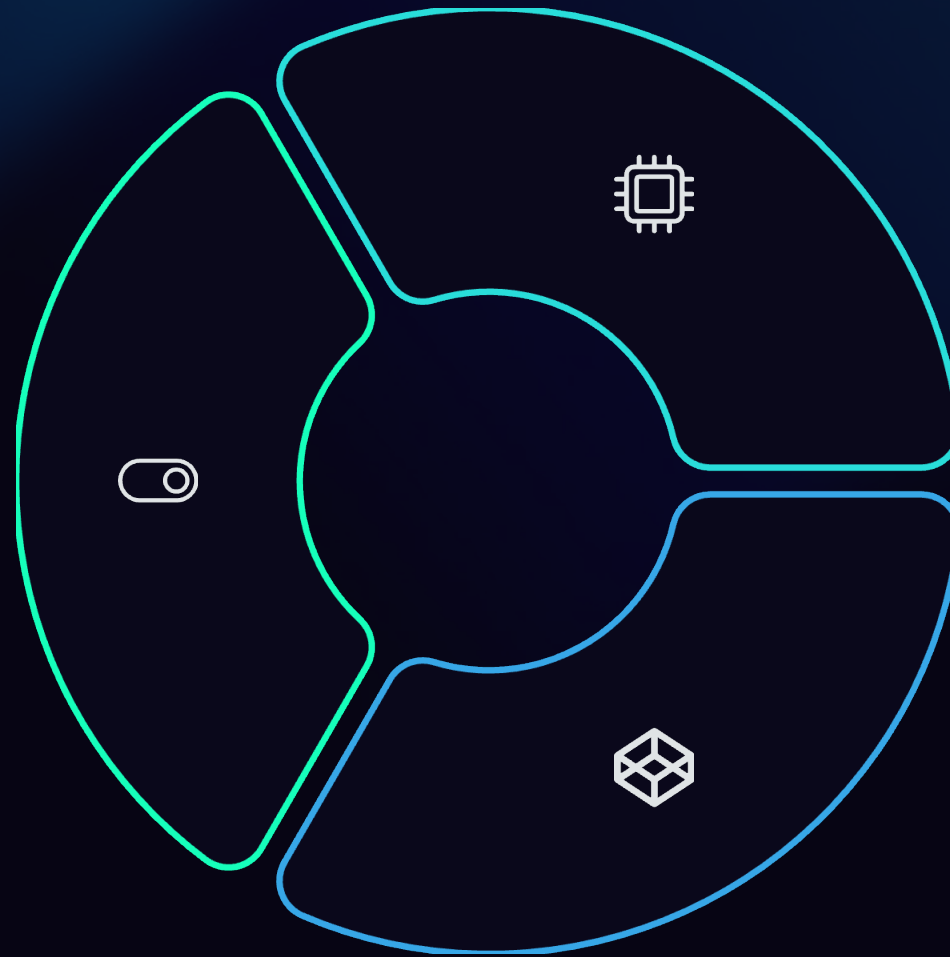
Labels & Aliases

Simplified code access



How Kconfig & DeviceTree Work Together

Kconfig
Enables drivers
(CONFIG_GPIO=y)



DeviceTree

Defines hardware (LED on
GPIO0, pin 13)

Application

Uses enabled, defined hardware



Live Example – LED

Blink

Start with Hello

World

Base project structure

Edit Files

prj.conf and main.c

Add app.overlay

Define LED alias

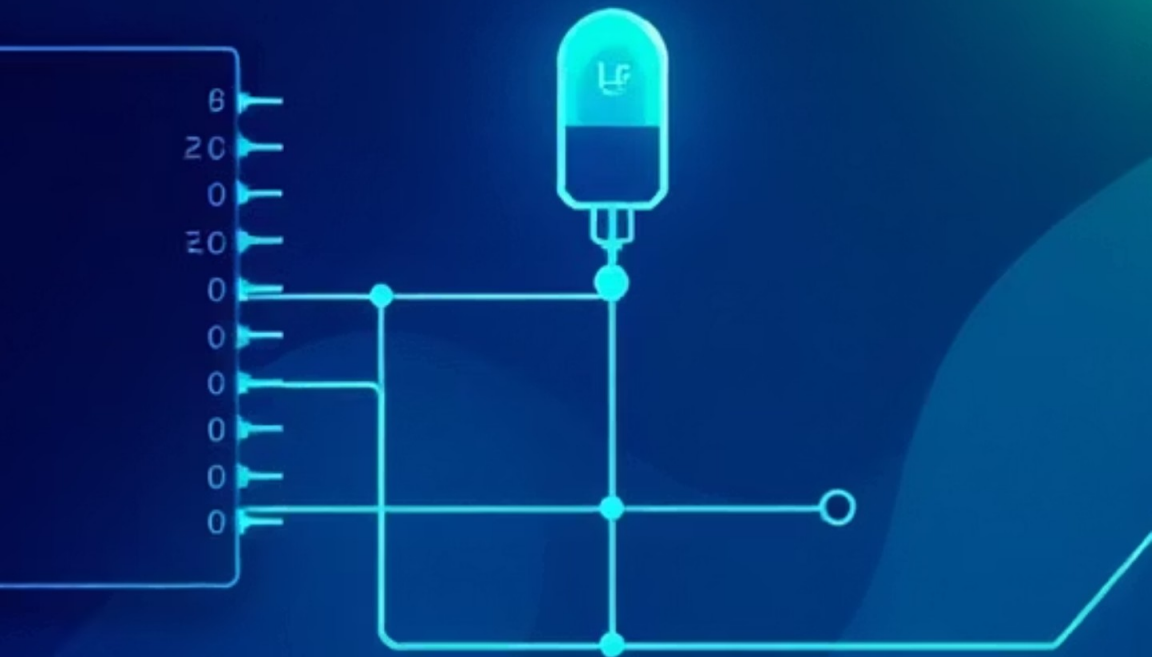
prj.conf – Example

```
CONFIG_GPIO=y
```

Enables the GPIO driver subsystem



```
1 emmeded systems
2
3 clunfntione (file (;
4   configuration for suruan );
5   contlao settings
6   configured etting;
7
8 / emmeded system emftigvation );
9 emmeded system:
10  cratelér {
11    rAnnedOwma*-Cesinn'calers'();
12
13    onetlr:tol({})
14    ine rfastu, =deturcan #ll/);
15    emmeded set:file;
16    corader file rorerill;
17
18    conttater, celltton system;
19    fileltioo({)
20    grestatinn; );
21    conctosifsl, file;
22    colected's entrrigut on );
23    coteradomient, eddelling);
24    recanne (ceen);
25  }
26  frlen file codaer (ald emmeded systems; setting ); {
27  );
28  }
29 }
```

app.overlay -

```
/ {  
    aliases {  
        led0 = &led0;  
    };  
};
```

Defines hardware reference for LED

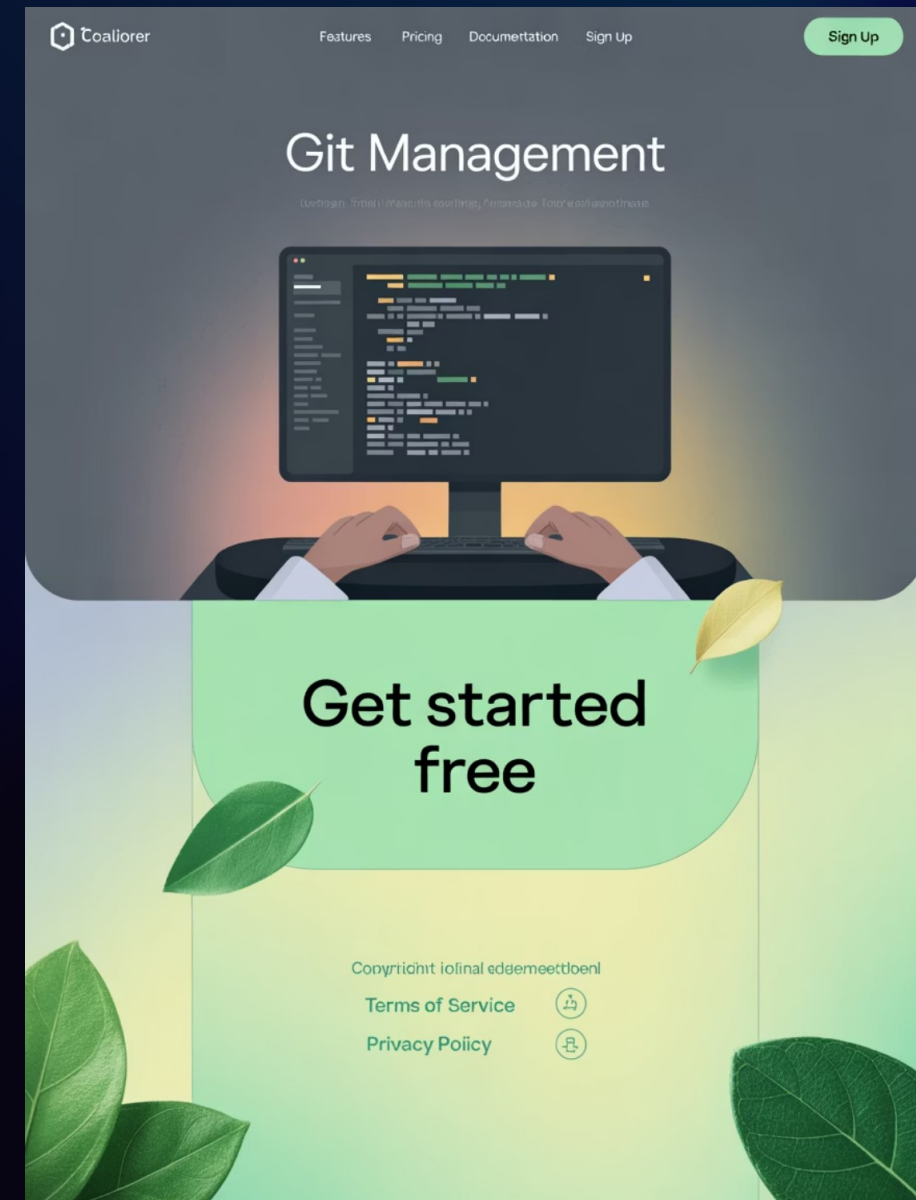
main.c – DeviceTree Usage

Add the code from blinky led sample to main.c



Run the sample

- build and flash on nrf52dk
files .dts and _defconfig from the folder
boards/nrf52dk generates the configuration
- Check the file build/zephyr/.config
- Check the file build/zephyr/zephyr.dts



Explore menuconfig

Run: `west build -t menuconfig`

Enable or disable options

Compare output with `build/zephyr/.config` now